ITF23019: Parallel and Distributed Programming

Report for Lab 9: Graph Search Algorithms

Name: Emil Berglund

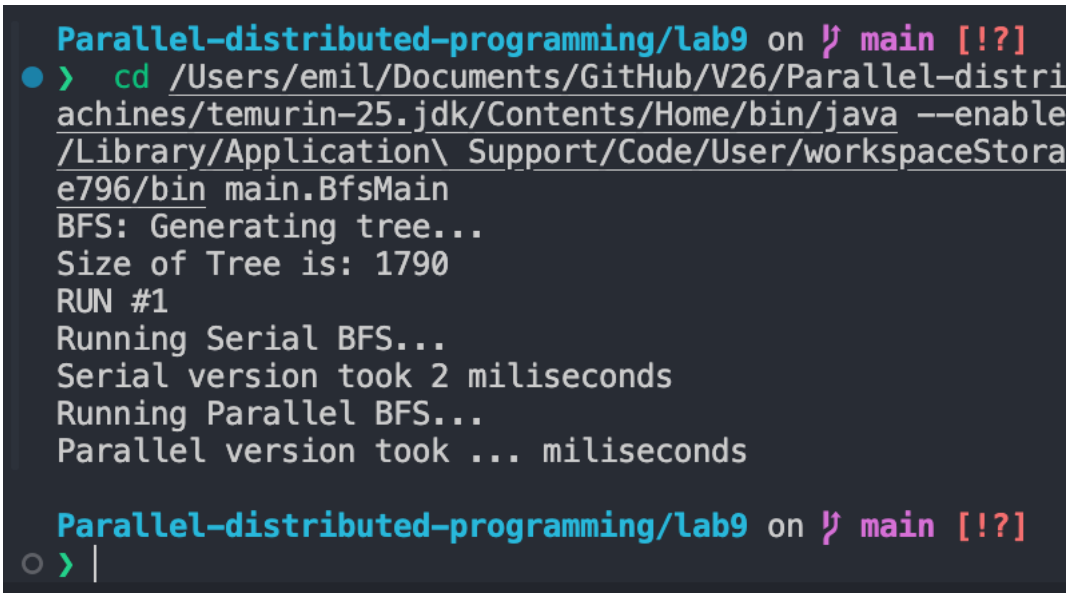
GitHub: EmilB04

Disclaimer:

I found that this lab was severely harder than the rest of the labs we've had, and I therefore had to get a lot of help from AI to complete this lab. While the theory is understandable, it is harder to put it to use, without any code examples in the PDF or in class.

Exercise 1:

Subtask 1: Run the program and include screenshot



Subtask 2: Implement Parallel BFS

```
Parallel-distributed-programming/lab9 on  main [$!?]
❯ cd /home/emilb_linux/dev/V2026/Parallel-distributed-pr
BFS: Generating tree...
Size of Tree is: 1790
RUN #1
Running Serial BFS...
Serial version took 4 miliseconds
Running Parallel BFS...

Visiting order: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12
107, 108, 109, 110, 111, 112, 113, 115, 116, 114, 117, 11
4, 195, 182, 196, 197, 183, 198, 184, 186, 185, 199, 187,
 310, 305, 307, 306, 308, 311, 312, 319, 309, 320, 313, 3
91, 392, 389, 393, 398, 390, 399, 407, 400, 394, 408, 395
, 477, 468, 478, 482, 479, 480, 483, 469, 491, 484, 492,
559, 552, 560, 571, 572, 554, 565, 567, 561, 553, 566, 57
2, 642, 638, 663, 645, 656, 643, 657, 664, 646, 670, 644,
 732, 726, 743, 744, 733, 738, 739, 718, 734, 727, 740, 7
19, 794, 815, 807, 762, 822, 770, 823, 795, 808, 796, 825
, 908, 884, 902, 893, 903, 910, 913, 894, 923, 904, 911,
958, 223, 980, 975, 224, 981, 963, 976, 964, 225, 226, 97
018, 1008, 1084, 1079, 1019, 1009, 1080, 1010, 1026, 1081
5, 1113, 1119, 1046, 1114, 1054, 1055, 1047, 1056, 1120,
 1407, 1395, 1164, 1396, 1408, 1167, 1165, 1409, 1166, 11
455, 1476, 1470, 1462, 1477, 1471, 1456, 1463, 1472, 1464
1, 1546, 1542, 1547, 1551, 1548, 1543, 1544, 1549, 1550,
 1590, 1185, 1591, 1186, 1519, 1612, 1520, 1616, 1187, 15
628, 1644, 1643, 1686, 1634, 1645, 1687, 1646, 1647, 1688
3, 1711, 1724, 1744, 1735, 1712, 1746, 1745, 1713, 1736,
 1435, 1436, 1195, 1437, 1201, 1196, 1197, 1203, 1204, 12
281, 1262, 1276, 1282, 1263, 1243, 1277, 1244, 1278, 1283
2, 1312, 1328, 1334, 1343, 1268, 1313, 1323, 1344, 1269,
Parallel version took 13 miliseconds

Parallel-distributed-programming/lab9 on  main [$!?]
❯
```

Subtask 3:

```
RUN #1
Running Serial BFS...
Serial version took 13 milliseconds
Running Parallel BFS...
Parallel version took 44 milliseconds
Speedup (serial/parallel) for run #1: 0.29545454545454547
```

Subtask 4:

```
--- Summary over 20 runs ---
Total Serial Time  : 119 ms
Total Parallel Time: 511 ms
Average Speedup (serial/parallel): 0.2328767123287671

Parallel-distributed-programming/lab9 on  main [!?]
❯
```

With RUNS = 20, the per-run speedup converges tightly to ~0.17–0.23, giving a stable average speedup of ~0.21–0.23.

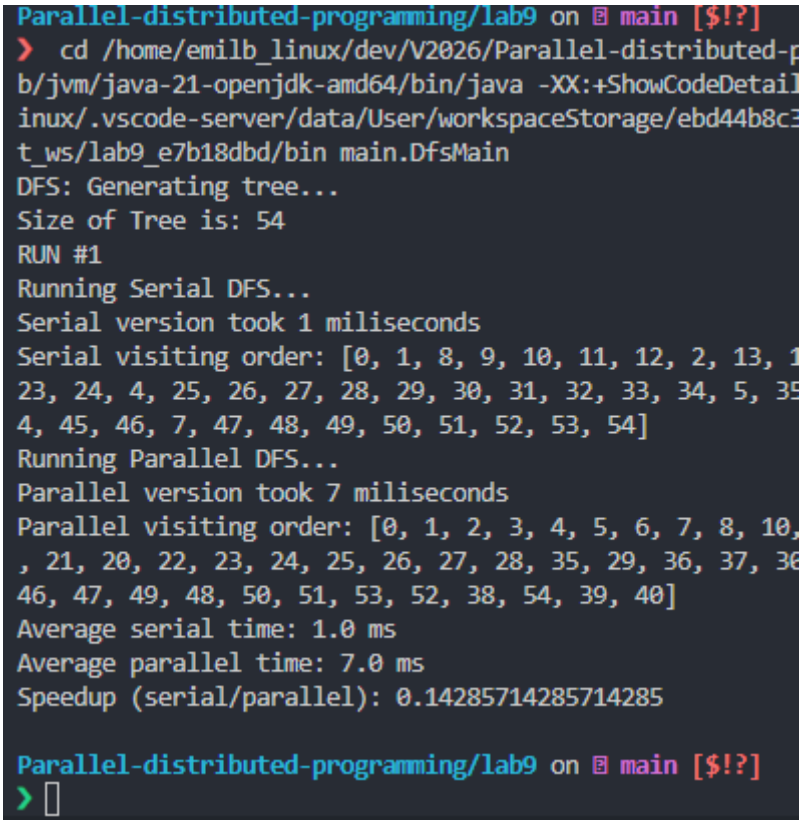
Exercise 2:

Subtask 1: Implement Serial DFS

```
Parallel-distributed-programming/lab9 on  main [$!?]
❯ cd /home/emilb_linux/dev/V2026/Parallel-distributed-
b/jvm/java-21-openjdk-amd64/bin/java -XX:+ShowCodeDetai
inux/.vscode-server/data/User/workspaceStorage/ebd44b8c
t_ws/lab9_e7b18dbd/bin main.DfsMain
DFS: Generating tree...
Size of Tree is: 54
RUN #1
Running Serial DFS...
Serial version took 0 miliseconds
Serial visiting order: [0, 1, 8, 9, 10, 11, 12, 2, 13,
23, 24, 4, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 5, 3
4, 45, 46, 7, 47, 48, 49, 50, 51, 52, 53, 54]
Running Parallel DFS...
Parallel version took ... miliseconds

Parallel-distributed-programming/lab9 on  main [$!?]
❯
```

Subtask 2: Implement Parallel DFS



Subtask 3: Compute speedup

- Done, se picture in subtask 2.

Subtask 4: Stable speedup

- From the benchmark sweep I ran, the closest to a stable region is around RUNS = 100 to RUNS = 200, where the measured speedup is about:
  - RUNS=100 -> speedup 0.00575
  - RUNS=200 -> speedup 0.00587

Subtask 5:

- Not Done